

London Bicycles

Data Engineering Insight & Risk Report

GCP Project: m2-dataset-study | Dataset: bigquery-public-data.london_bicycles | Generated: 19 May 2026

1. Executive Summary

This report analyses the Transport for London (TfL) Santander Cycles public dataset hosted on Google BigQuery. The dataset spans **83.4 million rental records** from January 2015 to January 2023 across **~800 docking stations**. The study surfaces actionable business insights, proposes a data warehouse architecture, and identifies the key risks that must be managed to deliver a reliable, production-grade data pipeline.

The central business problem is: *TfL operates one fleet serving two completely different markets — commuters and leisure/tourist riders — without optimising the fleet rebalancing strategy for either segment*. This mismatch causes predictable, preventable supply failures every day, directly reducing ridership revenue and customer satisfaction.

Metric	Value
Total rental records	83,434,866
Date range	4 Jan 2015 – 15 Jan 2023
Active stations (peak)	~800
Unique bikes ever tracked	~24,700
Avg trip duration (all)	21.9 minutes
Peak single month (Jul 2022)	1,303,376 rentals
Trough month (Jan 2022)	741,875 rentals
Missing end-station records	561,495 (0.67%)

2. Key Insights from the Data

2.1 Two Markets, One Fleet

Usage patterns differ fundamentally by day type, exposing two distinct user segments that share the same docking infrastructure:

Hour	Weekday Rentals	Weekend Rentals	Ratio W/E
08:00	7,156,970	412,928	17×
12:00–14:00	~5.7M (broad)	~5.8M (broad)	~1×
17:00–18:00	13,221,451	~3.1M	4×

- Weekday:** sharp bimodal spikes at 08:00 and 17–18:00 → commuters using bikes as last-mile transport to/from stations.

- **Weekend:** flat plateau 10:00–17:00 → leisure riders and tourists with no time pressure.

This single observation drives the entire rebalancing strategy: weekday mornings require bikes at residential/Tube-adjacent stations; weekends require bikes near parks and attractions.

2.2 Top Routes Are All Leisure Circuits

The 10 highest-volume routes are closed loops inside Hyde Park and Kensington Gardens. No point-to-point commuter route appears in the top 10, confirming that leisure tourism is a dominant revenue source.

Route (Start → End)	Trips	Avg Duration
Hyde Park Corner → Albert Gate	33,122	42.2 min
Hyde Park Corner → Triangle Car Park	31,747	29.7 min
Black Lion Gate → Palace Gate	30,908	23.5 min
Albert Gate → Hyde Park Corner	29,861	42.5 min
Triangle Car Park → Hyde Park Corner	29,786	30.0 min
Aquatic Centre → Podium (Olympic Park)	23,501	42.9 min

Implication: high-dock capacity at Hyde Park stations is not a luxury — it is operationally critical for the weekend leisure segment.

2.3 COVID Signature in Trip Duration (2020)

Average trip duration jumped from 19.5 min (2019) to 26.2 min (2020) — a 34% increase — before reverting to ~21 min in 2021. No other year shows such a deviation. This is consistent with lockdown-era behaviour: commuter trips disappeared, and people used bikes purely for outdoor exercise.

Year	Rentals	Avg Duration (min)	Unique Bikes
2018	10,441,884	19.9	12,934
2019	10,310,063	19.5	12,943
2020	10,287,040	26.2 ↑	14,743
2021	10,821,289	21.9	14,962
2022	11,027,519	21.3	24,715

The pipeline should include an anomaly-detection model that flags month-over-month duration deviations > 15% for investigation — this is a leading indicator of macro disruptions.

2.4 Seasonal Demand — 3× Summer vs Winter

Month (2022)	Rentals	Round Trips	Trips > 1hr
January	741,875	24,749	15,899
April	1,022,467	43,469	41,662
July	1,303,376	45,017	51,014
September	609,978	20,453	14,235
December	421,860	13,449	8,602

September 2022 is anomalously low (609k vs 835k in October), likely caused by restricted access near royal sites following the death of Queen Elizabeth II (8 September 2022). The pipeline should maintain a **calendar of**

external events to contextualise such drops.

2.5 E-Bike vs Classic — Different Behaviour Profiles

Bike Model	Rentals	Avg Duration	Unique Bikes	% of Fleet
NULL / Untagged	81,048,143	~21.9 min	~20,779	97.1%
CLASSIC	2,254,688	23.5 min	12,284	2.7%
PBSC_EBIKE	132,035	18.9 min	492	0.2%

E-bikes show shorter average durations (18.9 min vs 23.5 min for Classic) suggesting use by time-conscious commuters rather than leisure riders. However, 97% of records have no bike model tag — limiting fleet-level segmentation to the 2022–2023 window where tagging was introduced.

3. Proposed Data Pipeline Architecture

The architecture follows a modern ELT pattern entirely within BigQuery (EU region), using dbt Core for transformations and Python + SQLAlchemy for analysis. No data is copied out of the public dataset; all transformations create materialised views or tables inside m2-dataset-study.

Layer	Tool	Output
Source	bigquery-public-data (read-only)	Raw cycle_hire, cycle_stations
Staging	dbt — stg_cycle_hire, stg_stations	Cleaned, typed, filtered records
Dimensions	dbt — dim_date, dim_station, dim_bike	Conformed dimension tables
Fact	dbt — fact_rentals	83M row central fact table
Marts	dbt — mart_station_demand, mart_monthly	Aggregated analytical tables
Analysis	Python / pandas / SQLAlchemy	Jupyter notebooks & charts
Orchestration	GitHub Actions (cron) — optional	Scheduled dbt runs & tests
Quality	dbt tests + Great Expectations	Null / duplicate / range checks

3.1 Star Schema Design

The schema is optimised for the two primary analytical questions: *when is demand highest?* and *which stations are most imbalanced?*

Fact table: fact_rentals

rental_id, bike_id (FK), start_station_id (FK), end_station_id (FK),
date_id (FK), duration_sec, duration_min, is_round_trip,
trip_segment -- commute_am | commute_pm | leisure | night

Dimension tables:

dim_station (id, name, latitude, longitude, docks_count, is_temporary)
dim_bike (bike_id, bike_model, model_category)
dim_date (date_id, date, year, month, day_of_week,
season, is_weekend, is_uk_bank_holiday)

The **trip_segment** derived column is the key transformation: it labels each rental as commute_am (Mon–Fri 06:00–09:30), commute_pm (Mon–Fri 16:00–19:30), leisure, or night. This single column enables all dual-market analysis without repeated CASE logic in every query.

4. Risk Register & Mitigations

The following risks were identified from direct interrogation of the dataset. Each risk is rated by Likelihood (L) and Impact (I) on a 1–3 scale.

R1: Station Metadata Gap — 52% of hire stations have no dim_station record		Likelihood: High Impact: High
Finding	cycle_hire references 1,625 unique station IDs; cycle_stations contains only 785 records. 840 stations (52%) have no latitude, longitude, or dock-count — making geo-spatial and capacity analyses incomplete for over half the network.	
Mitigation	Enrich dim_station from the TfL Unified API (api.tfl.gov.uk/bikepoint) which returns live station metadata. Schedule a weekly dbt seed refresh. For historical records with no match, flag as 'unresolved_station' and exclude from geo queries — do not silently drop or NULL-fill with wrong data.	

R2: bike_model NULL for 97% of Records — Limits Fleet Segmentation		Likelihood: High Impact: Medium
Finding	Only 2,386,723 records (2.86%) carry a non-null, non-empty bike_model value. E-bike vs. Classic analysis is therefore only valid for 2022–2023 data. Applying model-based insights to the full 2015–2022 history would be misleading.	
Mitigation	Scope all bike_model analysis explicitly to 2022+. In dbt, create a model_category dimension that maps NULL/empty to 'Unknown' and document the coverage limitation in the mart. Suppress e-bike charts for pre-2022 time ranges.	

R3: Duration Outliers — 46,000+ Records Exceed 24 Hours		Likelihood: High Impact: Medium
Finding	3,227,082 trips (3.87%) last between 1 and 24 hours; 47,133 (0.056%) exceed 24 hours, with some reaching multiple days. These likely represent stolen or lost bikes. Unfiltered, they inflate average duration KPIs significantly.	
Mitigation	In stg_cycle_hire, apply a configurable duration cap (default: 86,400 sec / 24hrs) as a dbt variable. Flag over-cap records as is_anomalous_duration = TRUE and route them to a separate mart_anomalous_trips for fleet-loss reporting. Never silently discard — these records have operational value.	

R4: Missing End Stations — 561,495 Records (0.67%) Unresolvable		Likelihood: Medium Impact: Medium
Finding	561,495 trips have NULL end_station_id. These cannot be used for route, flow-balance, or round-trip analysis. Possible causes: bike returned outside a dock, system error at docking, or theft.	
Mitigation	Exclude from route and flow-balance marts. Include in volume KPIs (they are real rentals). Add a dbt test: accepted_values for end_station_id not null, with a warning — not error — threshold so the pipeline doesn't fail on expected noise.	

R5: EU Region Lock — All Resources Must Be in EU		Likelihood: Medium Impact: High
Finding	The london_bicycles dataset is stored in the BigQuery EU multi-region. Any dataset created in m2-dataset-study in US or another region will cause cross-region query failures with no automatic fallback.	
Mitigation	Set location: EU in dbt profiles.yml. Create all BigQuery datasets with --location=EU. Add a CI check (GitHub Actions) that validates the location of every dataset before running dbt. Document this as a hard constraint in the project README.	

R6: Read-Only Source — No Backfill or Correction Possible		Likelihood: Medium Impact: Medium
Finding	bigquery-public-data is a Google-managed public dataset. The project cannot correct errors, add missing records, or control data freshness. The dataset ends at January 2023 with no guarantee of updates.	
Mitigation	Design all dbt models to be idempotent and incremental from a known watermark. Document the data cutoff date prominently. For real-time needs, consider supplementing with the TfL live feed via Cloud Pub/Sub into a separate BigQuery table in m2-dataset-study.	

R7: External Event Contamination — Anomalies Silently Distort KPIs		Likelihood: Low Impact: Medium
Finding	September 2022 shows 609,978 rentals — 27% below October 2022 — due to area closures following the Queen's death. Without context, automated reports would flag this as a system failure or data pipeline error.	
Mitigation	Maintain a seed table (dbt seed: external_events.csv) with dated events (royal events, strikes, extreme weather). Join to mart_monthly_trends and add an is_disrupted_period flag. Alert consumers to exclude flagged months from baseline KPI calculations.	

R8: Zero and Negative Duration Records — Data Integrity Noise		Likelihood: Low Impact: Low
Finding	39,467 trips have zero duration and 238 have negative duration. Zero-duration trips may represent dock errors (bike docked and immediately re-rented); negative durations are clearly erroneous.	
Mitigation	In stg_cycle_hire, filter out duration <= 0 from all analytical models. Add a dbt singular test: assert no negative or zero durations in fact_rentals. Log count of filtered records per run for auditability.	

5. ELT Transformation Plan (dbt)

dbt Model	Type	Key Logic
stg_cycle_hire	View	Cast types; filter duration ≤ 0; flag is_anomalous_duration; flag missing end_station
stg_stations	View	Join cycle_stations + TfL API seed; flag unresolved stations
dim_date	Table	Generate date spine 2015–2023; add season, is_weekend, is_uk_bank_holiday
dim_station	Table	Deduplicate; add geo fields; mark temporary/removed stations
dim_bike	Table	Map NULL/empty bike_model → 'Unknown'; add model_category

dbt Model	Type	Key Logic
fact_rentals	Incremental	Join all dims; compute duration_min, is_round_trip, trip_segment
mart_station_demand	Table	Hourly departures & arrivals per station; net_flow = arrivals – departures
mart_monthly_trends	Table	Monthly KPIs; join external_events seed for is_disrupted_period
mart_route_volume	Table	Top routes by trip count; exclude missing end stations

6. Data Quality Testing Plan

Test	Tool	Threshold	Action on Fail
fact_rentals.duration_min > 0	dbt singular test	0 failures	Block pipeline
fact_rentals.rental_id unique + not null	dbt generic test	0 failures	Block pipeline
end_station_id null rate < 1%	dbt test	Warn at 0.7%, Error at 1%	Alert only
Monthly rental count within 3σ of mean	Great Expectations	Warn on breach	Alert + flag mart
avg duration change < 30% MoM	Great Expectations	Warn on breach	Alert + investigate
All fact station_ids exist in dim_station	dbt relationship	Warn (due to metadata gap)	Alert only
No trips with start_date > end_date	dbt singular test	0 failures	Block pipeline

7. Python Analysis — Recommended Notebook Structure

Notebook	Analysis	Business Question Answered
01_demand_patterns.ipynb	Heatmap: hour × day_type, rental counts	When should we rebalance?
02_station_flow.ipynb	Net flow per station (arrivals – departures)	Which stations empty/overflow?
03_seasonality.ipynb	Monthly trend + anomaly detection (3σ)	How should fleet size scale?
04_route_analysis.ipynb	Top routes, round-trip ratios by station	Where do leisure riders go?
05_ebike_impact.ipynb	E-bike vs Classic duration & station mix	Where should e-bikes be placed?

Connect via SQLAlchemy using the BigQuery dialect: `create_engine("bigquery://m2-dataset-study")`. Use pandas `read_sql` for mart queries — avoid pulling raw fact_rentals (83M rows) directly into memory. Always query the marts.

8. Executive Presentation Outline (10 min)

Slide	Title	Key Message	Time
1	The Problem	One fleet, two markets — neither optimised	1 min
2	The Evidence	Weekday vs Weekend demand shape (dual chart)	1.5 min
3	The Opportunity	3× seasonal swing + top leisure routes map	1 min

Slide	Title	Key Message	Time
4	Our Architecture	ELT pipeline: BigQuery → dbt → Python → insights	2 min
5	Key Findings	Station flow imbalance + COVID anomaly + e-bike signal	2 min
6	Risks & Controls	Top 3 risks (R1, R3, R5) + mitigations	1 min
7	Recommendations	Segment-aware rebalancing + e-bike placement model	1 min
8	Q&A;	Ready to answer	5 min

ROI framing for executives: Even a 1% improvement in bike availability during peak 08:00–09:30 slots (currently ~7.2M trips/year at that window) translates to ~72,000 additional rentals. At TfL's reported £2–3 average net revenue per rental, that is £144k–£216k incremental annual revenue from rebalancing optimisation alone.

9. Recommended Tech Stack & Justification

Component	Choice	Why
Cloud platform	GCP / BigQuery (EU)	Source data is already here; zero egress cost; serverless scale
Transformation	dbt Core	SQL-native; version-controlled; built-in testing; free tier
Orchestration	GitHub Actions (cron)	Free for public repos; no infra; sufficient for batch cadence
Analysis	Python + pandas + BigQuery connector	Assignment requirement; Jupyter-friendly
Data quality	dbt tests + Great Expectations	dbt for structural; GE for statistical anomaly detection
Diagramming	Excalidraw	Free; clean architecture diagrams; exportable to SVG/PNG
Documentation	dbt docs generate	Auto-generates data catalog from model YAML — zero extra work